



The physics of chemical strengthening of glass: Room for a new view

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ABSTRACT

Glasses strengthen during an ion exchange experiment as a result of the high surface compression that develops from the stuffing of the invading alkali ion into the smaller host alkali ion site and the non-relaxation of this compression. The kinetics are described in terms of a Nernst–Planck interdiffusion coefficient of the exchanging ions in a stack of mixed-alkali glass layers with varying ratios of the alkali. The interdiffusion coefficient is suppressed by compression and enhanced by tension. However, the measured surface compression is usually 2–4 times lower than that calculated using thermal stress analogy incorporating an elastic suppression of the molar volume difference of the as-melted mixed-alkali glasses. A new view of the physics of deformations and failure of stress to continue to buildup is presented in an effort to seek technology improvements. Reduced stresses are ascribed to network yield in two separate modes: yielding of the shape-conserving hydrostatic stress component which prevents elastic expansion of molar volume and yielding of the volume-conserving deviatoric (pure shear) stress which leads to measurable shape change. An anomalous subsurface tension is also explained in this view. By drawing analogy to microindentation, it is concluded that optimum network topology glasses should undergo the least plastic deformation and, hence, develop a higher surface compression.

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1. Introduction

The exchange of large alkali ions such as K^+ from a molten KNO_3 bath with comparatively smaller host alkali ions such as Na^+ in an alkali-containing glass at temperatures below the strain point of glass leaves the glass surface in a state of high compression. Because glass products usually break due to excessively applied tension acting on a surface flaw, the introduction of a high surface compression strengthens the glass. This process of ‘ion-exchange strengthening’ or ‘chemical strengthening’ was described by Kistler [1] and Aclouque and Tochon [2] in 1962. Relative to the thermal tempering process, chemical strengthening is advantageous in that the developed surface compression is usually much higher, that there is generally no measurable geometric distortion and that it can be applied to relatively thin and complex geometry products such as tubes quite readily. The slow nature of ionic diffusion, however, requires immersion in salt bath for several hours and, hence, is expensive. Several chemically strengthened glass products are successfully marketed: notable among these are the aircraft cockpit windshields, autoinjector cartridges to prevent anaphylactic shock, photocopier glass, and the thin display windows in personal communication electronic devices such as the cell phones and the MP3 players. After nearly five decades since the discovery, the markets appear to be ready for newer products which emphasize global efforts against terrorism, and for energy generation and conservation. Examples are

transparent armor, glass substrates for harvesting solar energy, and high strength-to-weight ratio glass containers.

The chemical strengthening science and technology has been reviewed by several authors [3–7]. Mathematical principles of stress generation from concentration gradients were developed by Richmond et al. [8] for nitrogenation of α -iron using analogy to thermal stresses and subsequently adopted by Cooper and Krohn [9] to glass chemical strengthening. The stress field at location x in a thin semi-infinite (in y - and z -dimensions) glass plate of thickness $2L$ is given by (see Fig. 1):

$$\begin{aligned} (\sigma_{xx})_x &= 0 \\ (\sigma_{zz})_x &= (\sigma_{yy})_x = -\frac{BEC_x}{(1-\nu)} + \frac{1}{L} \int_0^L \frac{BEC_x}{(1-\nu)} dx \end{aligned} \quad (1)$$

where x is the diffusion direction, E = Young's modulus; ν = Poisson's ratio; and $B = (1/3V) (\partial V/\partial C)$ = linear network dilatation coefficient (called the “Cooper coefficient” in honor of the late Alfred R. Cooper). Like the linear thermal expansion coefficient α , the linear network dilatation coefficient is 1/3rd of the relative change in the volume V per unit exchange of concentration C . Essentially, the stress is obtained by replacing temperature T by concentration C , and the linear thermal expansion coefficient α by the linear network dilatation coefficient B .

After assuming $(\sigma_{xx})_x = 0$ as above, the strains at location x are given by:

$$(\varepsilon_{zx})_x = (\varepsilon_{yx})_x = 0 \quad (2)$$

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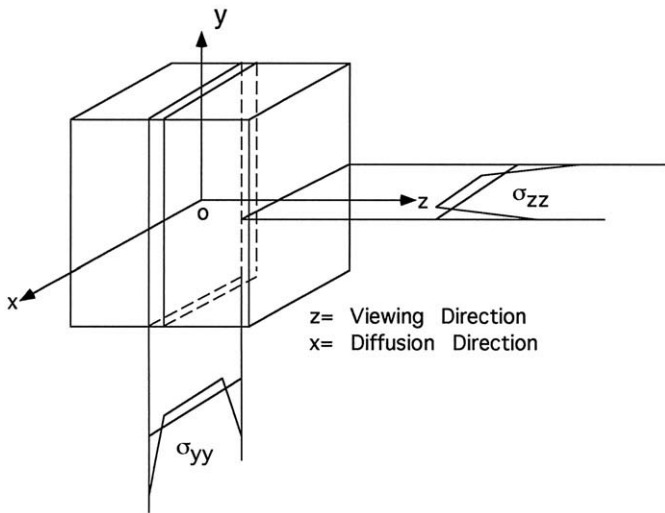


Fig. 1. Coordinate axes for diffusion and stress development during ion exchange.

$$(\epsilon_{xx})_x = -\frac{\nu}{E} [(\sigma_{yy})_x + (\sigma_{zz})_x] + BC_x = -\frac{2\nu}{E} [(\sigma_{yy})_x] + BC_x \quad (3)$$

$$(\epsilon_{zz})_x = (\epsilon_{yy})_x = \frac{1-\nu}{E} (\sigma_{zz})_x + BC_x. \quad (4)$$

Since $\int_0^L (\sigma_{yy})_x dx = \int_0^L (\sigma_{zz})_x dx = 0$, the average elastic strain in y- and z-directions is obtained by integrating Eq. (4) over the width of the specimen, and is $\frac{1}{L} \int_0^L BC_x dx$. Likewise, the net expansion in the x-direction from Eq. (3) is the free expansion, $\int_0^L BC_x dx$.

An example of a concentration profile and a corresponding (measured) stress profile in a sodium borosilicate glass is shown in

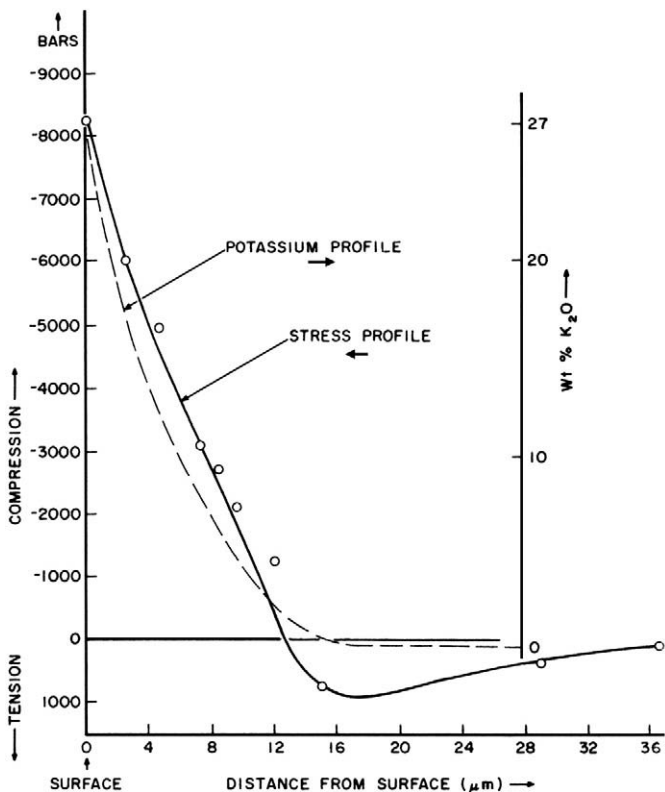


Fig. 2. Correspondence of stress and concentration profile as a function of depth in a sodium borosilicate glass. (Note 10 bars = 1 MPa). After Ref. [10].

Fig. 2, from Ref. [10]. Whereas there appears to be a general agreement between the stress and concentration, there are three primary discrepancies:

- (i) Not noted readily is the realization that the compressive stress magnitude computed using Eq. (1) and values of B for corresponding as-melted mixed-alkali composition glasses are roughly a factor of 2 to 4 too high relative to those measured. For $\text{Na}^+ \rightleftharpoons \text{K}^+$ exchange, the value of Cooper coefficient is calculated to be roughly $1.3 \times 10^{-3} (\text{mol}\% \text{K}_2\text{O})^{-1}$. However, experimentally, the observed stress profiles generally yield values of $3\text{--}6 \times 10^{-4} (\text{mol}\% \text{K}_2\text{O})^{-1}$.
- (ii) An anomalous tension maximum not predicted by Eq. (1) exists.
- (iii) A subsurface compression maximum exists [11] despite a monotonic decay of the invading ion concentration profile, see Fig. 3 taken from Sane and Cooper [12]. Whereas it is logical to assume that stress generation near the glass transition range must compete (successfully for strengthening) against relaxation of the same stress by viscous flow, Sane and Cooper were NOT able to account for all of the surface stress distribution by incorporating reasonable rates of viscous flow (assuming that the mixed-alkali layers in the surface had significantly lower glass transition temperatures).

Two other topics in the physics of chemical strengthening are dimensional swelling and fracture pattern. The stuffing of a large ion into a relatively smaller ion interstice will cause dimensional swelling of glass. In thin-walled products such as thin display windows in personal communication devices, such dimensional changes need to be considered during manufacturing. Likewise, although normal thick-walled chemically strengthened products do not have much interior tension and can often be sawed after strengthening without the risk of fracture, the magnitude of the interior tension in thin-walled chemically strengthened glasses can be high enough to cause the glass to frange into small pieces, often submillimeter-sized [13,14].

Of the characteristics describing the physics of chemical strengthening introduced above, physical swelling and frangibility are being left for a later date. This review focuses on the kinetics of diffusion and stress generation which are of paramount importance. In particular, I intend to present an improved model of glass network plasticity, the “new view”, which rationalizes the compressive stress reduction as well as the anomalous subsurface tension maximum. My aim is to seek guidance for glass composition development in an effort to obtain higher and higher surface compression more rapidly for improving the technology.

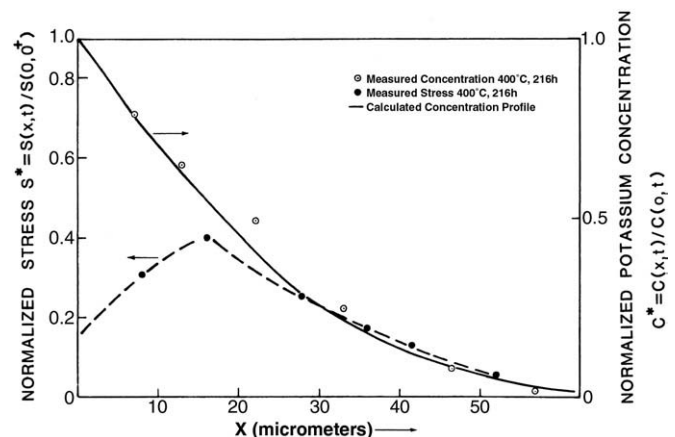


Fig. 3. Subsurface compression maximum in ion-exchanged soda lime silicates. After Ref. [12].

2. Kinetics of interdiffusion during chemical strengthening

This topic is quite well understood [15]. The concentration profile of the invading ion, hence that of the host ion (from an ion-for-ion exchange calculation) is described by an interdiffusion coefficient $\bar{D}$, and given by the Nernst–Planck expression (ignoring second order effects of thermodynamic activity):

$$\bar{D} = \frac{D_{Na}D_K}{D_{Na}N_{Na} + D_KN_K}, \quad (5)$$

where N_i is the fractional concentration of alkali ion i and D_i is its self diffusion coefficient in mixed-alkali glass compositions, and not in the parent glass. Since the self diffusion coefficients change over several magnitudes in mixed-alkali glass compositions (a manifestation of the familiar “mixed-alkali effect”), the interdiffusion coefficient is essentially controlled by the slower moving specie (see Fig. 4 taken from Ref. [10]). This unfortunately slows down the strengthening process (when conducted without electric or sonic field assistance). Misfortune does not stop here. It is now well-recognized that the interdiffusion coefficient is impeded by surface compression and accelerated by the interior tension. Experimental support for this hypothesis is provided by Varshneya and Dumais [16], and by Tomozawa and co-workers [17]. The effect of stress is to expand or constrict an average interstitial ion site [18]. A modification to the D_K due to stresses within the ion exchange region is proposed by Ingram et al. [19] and called the “anomalous mixed-alkali effect”. The lowered interdiffusion through the surface lowers the intake of the invading ions into the glass.

3. Stress generation and relaxation

At this point, it should be recognized that the stack of mixed-alkali glass layer structures in the ion exchange region do not resemble the glass structures of the corresponding as-melted compositions; they are in reality stuffed derivatives of the parent glass structure. Let us focus on the factor of 2 to 4 disparity between the observed stress magnitudes and those calculated using Eq. (1) and start with the premise that viscoelasticity of the exchanged structure can only partly

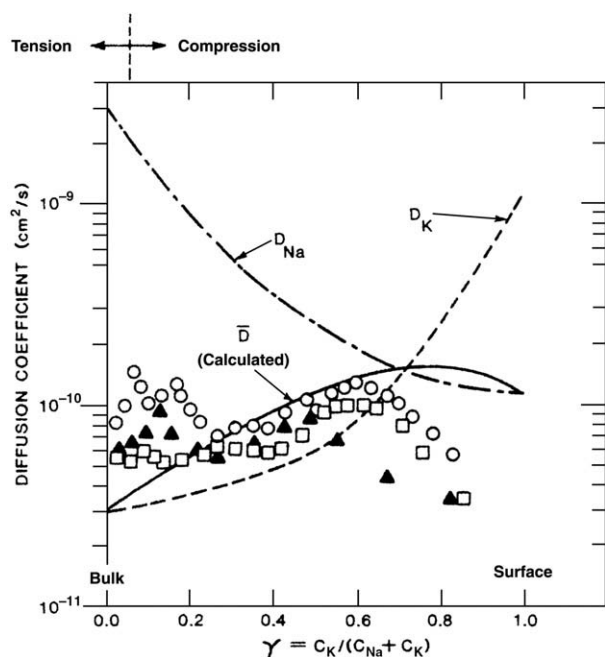


Fig. 4. Variation of the interdiffusion coefficient as a function of the mixed-alkali ratio covering the ion-exchanged region. After Ref. [10].

explain the stress reduction; there must be other low temperature mechanisms which prevent stress buildup. One such mechanism is plasticity. Early suggestions of network plasticity were made by Hale [20], Garfinkel and King [21], Varshneya [18], Jain and Varshneya [22] and Glebov et al. [23]. One envisions that, during stuffing, there will be an incompatible elastic expansion of the mesoscopic region surrounding an exchanged interstitial site. In a semi-infinite plate sample, the continuum comprised of many of such regions will be in the form of layers parallel to the surface. The enforcement of the elastic compatibility criteria will then develop a stress profile which is compression in the surface layers and tension in the interior layers.

Near T_g , the cations may themselves lead to stress reduction by causing local bonds to break and rearrange through *cation-induced relaxation of network*, or “CIRON” as suggested by Ingram et al. [19], see Fig. 5, eventually leading to a fully relaxed structure which is statistically identical to an as-melted glass. Note that large-scale cooperative bond-breaking and rearranging as in viscous flow is not invoked by the authors.

The “new view” that I like to present begins with the premise that CIRON itself need not be invoked. In the mesoscopic region around an exchange site, the bonds need only be bent or stretched beyond their elastic limit to provide partial, but permanent, accommodation for an invading larger ion. A stress tensor is composed of two components: the hydrostatic (dilatational) which changes the volume but conserves the shape, and a deviatoric which changes the shape and conserves the volume. Each of the components has an *independent* elastic limit α_{yh} and α_{ys} , respectively. *Each limit may be exceeded non-interactively even for a single K^+ stuffed in a single Na^+ site.* Bonds around K^+ stretch and bend

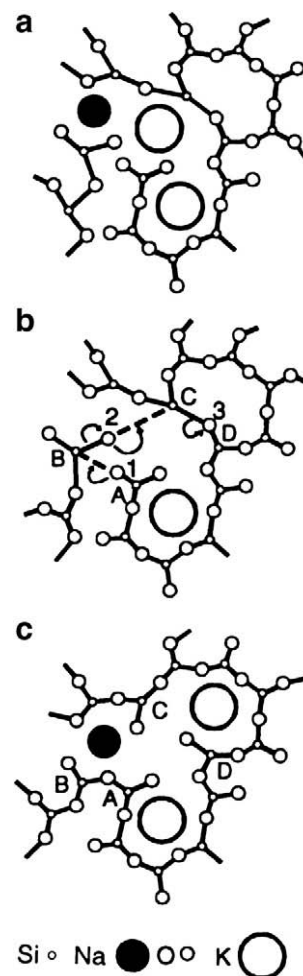


Fig. 5. Cation-induced relaxation of network (“CIRON”) according to Ingram et al. [19].

permanently. There is an incompatible strain developed in the mesoscopic region around the cation site which is partially elastic and partially plastic. Stress buildup corresponds only to the elastic strain, hence, is considerably lower than what Eq. (1) would predict. Fig. 6 shows in 2d

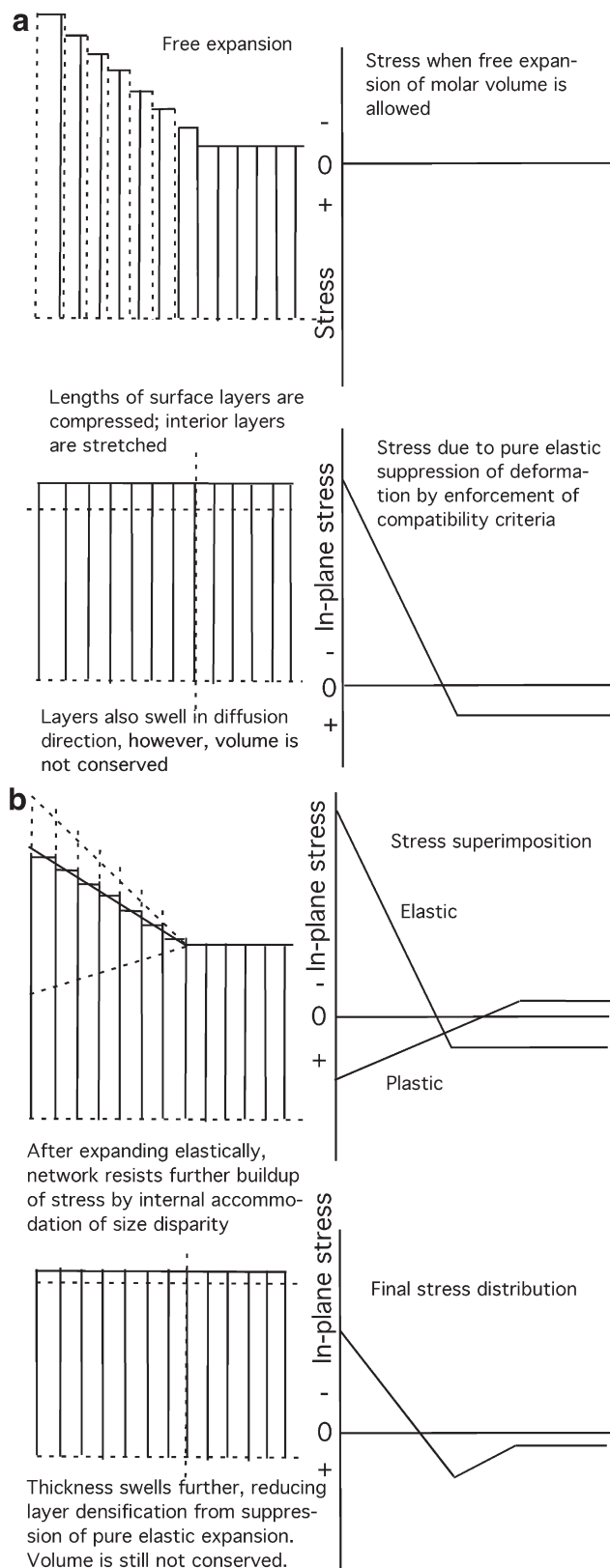


Fig. 6. a. Dimensional changes expected from “free” (virtual) elastic expansion and resulting stress from its suppression. b. Effect of network plasticity to reduce the stress buildup.

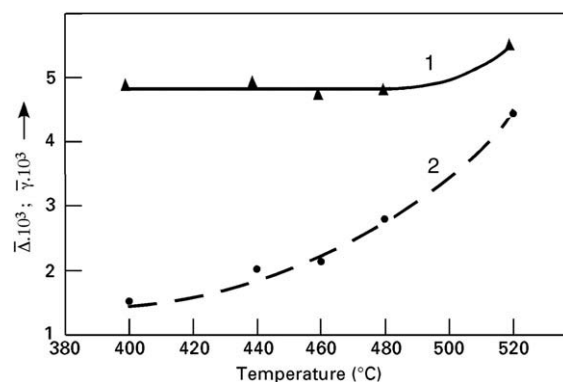


Fig. 7. Analysis of swelling as average relative dimensional changes in diffusion direction in ion-exchanged glass as a function of temperature. Curve 1 corresponds to estimated elastic “free” expansion. Curve 2 corresponds to estimated contribution from plasticity. Diffusion time = 4 h. Redrawn from Ref. [23].

schematic how glass layers expand on a continuum (macroscopic) scale after diffusion in the x -direction. In (a), the surface layers are shown to have “free” (virtual) expansion in *all* directions corresponding to the molar volume change of as-melted glasses. The enforcement of compatibility criteria develops surface compression and interior tension corresponding to a value of B given by the Cooper approach. Shown in (b) is the reduced macroscopic deformation in the y - (and z -) direction caused by internal accommodation of the size disparity. It would be termed plasticity since the accommodation implies achievement of some local potential energy minima for ions making it difficult for deformation to be recovered. The yielding of the dilatational stress component beyond its yield strength σ_{yh} will result in y - and z -dimensions being intermediate between that of fully constrained site and one where the deformations were totally elastic. Yielding beyond σ_{ys} will change shape of the y - and z -layers but will conserve the volume. Contrary to the principles of continuum mechanics, the mesoscopic regions except on the free surface can yield in all the directions, including the x -direction (diffusion). Swelling in the x -direction, hence, will not correspond to the “free” expansion in x . The relaxation of the hydrostatic component of stress will result in density changes as a function of x from those expected from suppression of pure elastic deformation. Such density changes are experimentally observed [24]. Note, net volume change during plasticity is not zero.

In this context, it is interesting to examine Glebov et al.’s analysis [23] of swelling in an ion-exchanged glass. They suggest that the incompatible expansion per unit length in the y - and z -directions¹ would be $(1 + \Delta - \gamma)$, where $\Delta = BC_x$ is the “free” expansion and γ is the contribution due to plasticity; note $\gamma < \Delta$. In the diffusion direction (x -axis), the expansion would be $(1 + \Delta + 2\gamma)$ based upon the premise that plasticity contribution was volume-conserving. Glebov et al. measured the σ_{yy} stress profile and the swelling of the x -dimension to compute Δ and γ . Average values of Δ and γ in a 4h ion exchange as a function of exchange temperature are shown in Fig. 7. Note the average “free” expansion remains relatively constant at low temperatures whereas the plastic contribution increases rapidly. A maximum value of Δ at low exchange temperatures was determined to be 0.01 compared to 0.018 from partial molar volumes. They ascribed the disparity to an unrelaxed structure by virtue of the K^+ ions occupying Na^+ sites which gradually jumped by thermal activation upon an annealing treatment to occupy their regular sites corresponding to an equilibrium-melted structure. The reader may note that Glebov et al. postulated the existence of a “CIRON” mechanism some 11 years prior to Ingram et al.

¹ Reader should kindly note that Glebov et al. use z -direction as the diffusion direction.

A more rational way of reconciling the disparity is that γ is not a volume-conserving shear plastic deformation γ_s alone; it has a hydrostatic plasticity component γ_h also. Thus, the elastic strain in the y - and z -directions at each x is $[1 + BC_x - \gamma_h - \gamma_s]$; that in the x -direction is $[1 + BC_x - \gamma_h + 2\gamma_s]$, where $\gamma = f(C_x)$. Separation of the average elastic and plastic contributions is, therefore, not possible from the two measurements made by Glebov et al.; one would need to measure the (i) σ_{yy} , (ii) swelling in the x -direction or the y -direction, and (iii) molar volumes of the as-melted glass compositions. Because of the likely x -dependence of γ , an analytical solution for compatible strain and stresses is not readily tractable. Instead, one may need to resort to finite element methods where the two yield strengths have been prescribed for each element.

The effect of plasticity is to superimpose a reverse stress distribution corresponding to the unexpanded portion of the free expansion (see Fig. 6b). The algebraic addition of the two (after taking some liberties) not only reduces the surface compression, but also generates the anomalous tension maximum typically observed in soda lime glasses. The reverse stress distribution would remain due to residual plastic deformation of the system if the ions were back-exchanged out somehow (assuming that the network remains rigid). Such a result would be unexpected from Glebov et al.'s argument. I submit that the experimental observations of Sglavo and Green [25] showing the appearance of tensile stress in the surface after back-exchanging result more likely from the residual reverse stress distribution of the permanent mesoscopic deformations and not due to the high thermal contraction coefficients of the surface layers as argued by the authors.

Schaeffer and Heinze [26] suggested that the anomalous tension maximum was a result of birefringence measuring ($\sigma_{yy} - \sigma_{xx}$) and not σ_{yy} ; the σ_{xx} (as per Eq. (1)) is zero only at the extreme outside surface, but, is non-zero subsurface in most finitely sized specimens. Using finite element simulation, Varshneya and Petti [27] showed that the entire magnitude of tension maximum could not be explained by a non-zero σ_{xx} . In yet another elegant and thought-provoking argument, Sane and Cooper [28] suggested that the anomalous tension maximum below the compression was an experimental artifact of slicing. I submit that the anomalous tension maximum may be *additionally* explained in terms of the effect of a plastic deformation of the diffusion layers as above.

An interesting question to ask is whether cooling the specimen down from exchange temperature to room temperature has any effect on the stress distribution. Some reduction in the stress magnitudes was shown quite convincingly in the experiments of Tyagi and Varshneya [29] who conducted $\text{Li}^+ \rightleftharpoons \text{Na}^+$ exchange in a lithium aluminosilicate glass plate placed immersed in NaNO_3 within a heated-stage silica cell and

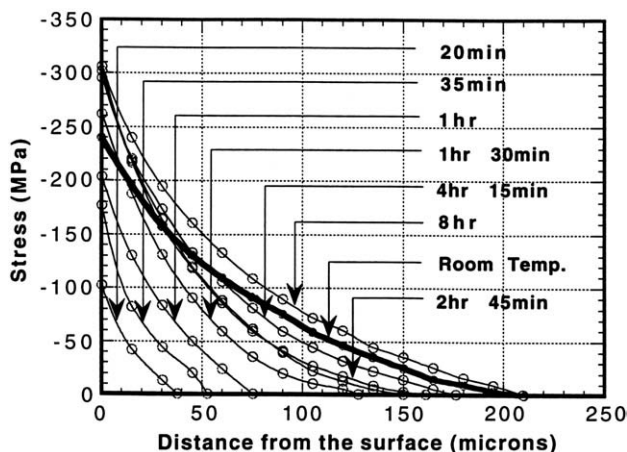


Fig. 8. Progressive stress development with time at 375 °C in a lithium aluminosilicate glass and loss of beneficial compression upon cooling to room temperature. After Ref. [29].

examined for optical birefringence progressively under a polarizing microscope at temperature. Their results are summarized in Fig. 8. Notice the gradual buildup of the compressive stress profile with time at an exchange temperature of 375 °C and an approximately 25% reduction in surface compression upon cooling to room temperature. Rapid heating to near the original exchange temperature restored the original stress magnitudes to within the measurement error. The reduction in stress due to cooling corresponded to a thermal contraction coefficient as high as $\sim 140 \times 10^{-7}/^\circ\text{C}$ on the surface relative to the $\sim 90 \times 10^{-7}/^\circ\text{C}$ of all the as-melted mixed-alkali compositions. A rational explanation for the apparently high surface thermal contraction coefficients remains illusive to-date.

The analogy of ion exchange to indentation should be apparent. An indenter is like a big ion. As the indenter is pushed into glass, the network densifies as well as shears (to create mounds or pile-up around the indentation rim). The unloading of the indenter creates a tensile stress field near the surface and at the base causing cracks to develop. The concept of the two network yield strengths has been adopted by Kermouche et al. [30] recently to estimate the densification around a microindentation using finite element analysis. According to them, the densification map as measured by Raman spectroscopy fits $\sigma_{yh} = 11.5$ GPa and $\sigma_{ys} = 6.5$ GPa. Unfortunately, continuum mechanics incorporating some yield criterion combining the two are not applicable to simulating ion exchange which requires stress calculations on an atomic scale. Molecular dynamics (MD) simulations employed by Keiffer [31] and Kob et al. [32] utilize the application of a hydrostatic pressure to compute irreversible densification in silica glass. One may argue that the external pressure at the onset of irreversible densification does not equate to hydrostatic yield strength of ionic bonds. One needs to perform actual ion exchange MD simulation and compute the stresses developing in a mesoscopic region extending subsequently to continuum using a multiscale approach.

My final and more important argument as a “new view” is to suggest that the hydrostatic yield strength σ_{yh} and the deviatoric yield strength σ_{ys} depend upon glass network topology. For deformation and cracking due to indentation in chalcogenide glasses, I have suggested [33] that the deviatoric yield strength achieves a maximum in glass with optimized network connections at an average covalent coordination $\langle r \rangle = 2.40$, see Fig. 9, in conformance with Phillips [34] and Thorpe's [35] topological constraint arguments. Thus, in oxide glasses where the network connections are optimized, presumably having minimized fraction of nonbridging oxygen, the deviatoric yield strength ought to be maximized. This would imply that there is less shear yielding during deformation from size disparity. Such glasses should ion-exchange strengthen better. Conversely, high alkali-containing glasses may have lower σ_{ys} simply because of lower connectivity, hence, may plastically shear more.

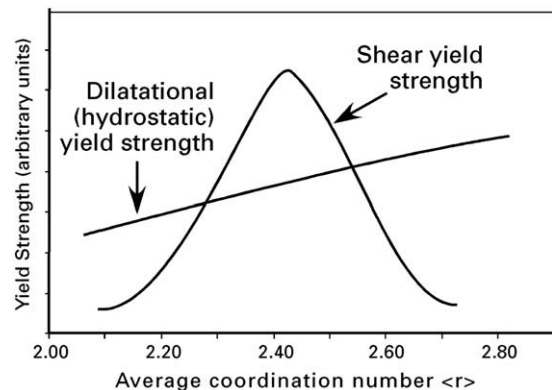


Fig. 9. Proposed dependence of hydrostatic yield strength σ_{yh} and deviatoric yield strength σ_{ys} upon network connectivity parameter $\langle r \rangle$. After Ref. [33].

4. Conclusions

This paper discusses two items concerning the physics of chemical strengthening of glass: (a) kinetics of ion exchange and (b) principles of stress generation and relaxation. Following conclusions are made:

- (1) The kinetics of ion exchange are described in terms of an interdiffusion coefficient that is given by Nernst–Planck expression and modified by the presence of macroscopic stress. It is argued that the average ion site size is decreased by compression, hence, the interdiffusion coefficient is reduced.
- (2) Although Cooper's analogy to thermal stress situation should apply, anomalies exist. Most puzzling has been a factor of 2–4 lower observed compression compared to that calculated from free volume changes of the mixed-alkali as-melted glass compositions.
- (3) Glass in the exchanged region is a stuffed derivative of the parent. Some portion of the size difference is accommodated plastically. This work emphasizes upon the separation of dilatational or hydrostatic component of stress from the deviatoric or pure shear component, either of which can exceed its corresponding yield strength to result in lowered beneficial stress magnitudes.
- (4) The anomalous tension maximum is explained, in part, by artifact of slicing, artifact of the non-zero stress in the diffusion direction subsurface, and by network plasticity.
- (5) The observed subsurface compression maximum may be a result of a combination of lowered surface T_g (viscoelastic relaxation and "CIRON"), a high apparent thermal contraction coefficient of the stuffed surface layers, and network plasticity.
- (6) By analogy to indentation experiments, it is suggested that glasses which have a well-connected network topology should have relatively high deviatoric yield strengths, hence, should strengthen better.

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